



ARTIFICIAL INTELLIGENCE · LECTURE 01

Introduction to AI

What AI Really Means · Classical vs Modern AI · Agents · History



■ WHERE WE'RE HEADED

Learning Objectives

01

Understand what “AI” really means — in four simple ways.

02

Learn the major AI categories, branches, and the classical vs modern AI view.

03

Understand the Turing Test — a famous way to check if a machine “seems” smart.

04

Learn a simple 4-part checklist (PEAS) to describe any AI system.

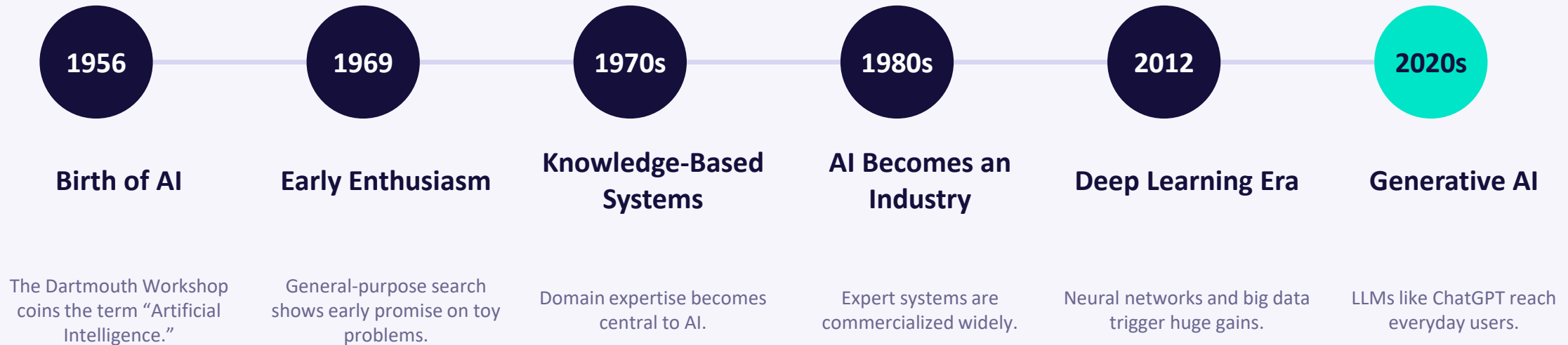
05

Learn the five main types of intelligent agents in AIMA.

06

See how AI has grown and changed since 1956.

The History of Artificial Intelligence



What Is AI? Four Simple Ways to Think About It

Nobody agrees on one single definition of AI — but here are the four main ideas people use.

Thinking Humanly

Model human thinking

Study how humans think, then build machines with similar problem-solving behavior.

Thinking Rationally

Use correct reasoning

Make a computer reason correctly using formal logic and valid conclusions.

Acting Humanly

Behave like a human

Make a computer communicate and behave so naturally that people may not distinguish it from a human.

Acting Rationally

Act to achieve the best outcome

Choose the action expected to achieve the best outcome in a given situation. This rational-agent view guides this course.

Three Capability-Based Categories of AI

How general can an AI system be across different tasks?

ANI

Narrow AI

AI that performs one specific task well — like spam filtering, recommendations, or voice assistance. This is the AI we use today.

WHERE WE ARE TODAY

AGI

General AI

AI that could learn and perform a broad range of intellectual tasks like a human. It remains a research goal — not current reality.

DOESN'T EXIST YET

ASI

Super AI

A hypothetical AI that would exceed human capability across most domains. It is a theoretical and debated idea.

PURE THEORY

Classical AI vs. Modern AI

Two broad approaches you will meet throughout this course.

Classical AI

Rules · Search · Logic · Reasoning

- Humans write knowledge and rules
- System searches possible actions
- Good for clear, structured problems
- Examples: expert systems, chess search

Modern AI

Data · Learning · Neural Networks · GenAI

- Model learns patterns from data
- Performance improves with examples
- Good for images, text, speech, prediction
- Examples: ML, DL, NLP, Computer Vision

Teacher line: Classical AI tells the machine the rules. Modern AI lets the machine learn patterns from data.

Branches of AI — The Main Job Types

Different “jobs” that AI can specialize in



Machine Learning

Teaching a computer by showing it lots of examples, instead of writing exact rules. (e.g., Netflix picks)



Natural Language Processing

Helping computers understand and talk in human language. (e.g., ChatGPT, Siri)



Computer Vision

Helping computers “see” and understand photos and videos. (e.g., Face unlock)



Robotics

Combining AI with physical machines that can move and do tasks. (e.g., Factory robots)



Expert Systems

A computer that follows rules like a human expert would. (e.g., Medical checkup tools)



Fuzzy Logic

Letting computers handle “maybe” answers, not just yes/no. (e.g., Smart AC temperature)



“Can machines think?”

— Alan Turing, 1950



A FAMOUS TEST

The Turing Test

Imagine texting two hidden “people” — one is a real human, one is a computer. If you can't tell which is which, the computer passes the test!

WHAT THE COMPUTER MUST BE ABLE TO DO

Understand Language (NLP)

so it can “talk” properly

Represent Knowledge (KR)

so it can store facts and context

Reason & Answer

so it can draw conclusions

Learn & Improve (ML)

so it gets smarter over time

A computer that can pass this AND see images AND move around in the real world passes the “Total” version of the test.

PEAS — A Checklist for Any AI

A simple 4-question checklist to describe how any smart agent works.



Performance Measure

How do we measure success or failure?



Environment

Where does it operate?



Actuators

How does it actually DO things?



Sensors

How does it “see” or “notice” things?



EXAMPLE

A Self-Driving Taxi

- P** Be safe, fast, and comfortable
- E** Roads, traffic, people
- A** Steering, brakes, signals
- S** Cameras, GPS, sensors



What Kind of “World” Is the AI In?

Can it see everything?

Fully Observable vs. Partially Observable

Chess: you see the whole board. Driving: you can't see around every corner.

Is the result guaranteed?

Deterministic vs. Stochastic

A light switch: press it, light turns on, guaranteed. A dice roll: you can never be sure.

Do past choices matter?

Episodic vs. Sequential

Checking one email for spam: stands alone. A chess move: affects the whole game ahead.

What Kind of “World” Is the AI In?

3 more questions — answered for our Self-Driving Taxi example:

Does the world wait for you?

Static vs. Dynamic

- *Dynamic — traffic keeps moving while the taxi “thinks.”*

Countable steps, or smooth flow?

Discrete vs. Continuous

- *Continuous — speed and position can be any number.*

Is it alone, or with others?

Single-Agent vs. Multi-Agent

- *Multi-Agent — other drivers and people must be considered.*

Five Types of Intelligent Agents

From simple reactions to systems that improve through experience



01 Simple Reflex

Reacts instantly to what it sees right now — no memory at all. (Like a knee-jerk reflex)



02 Model-Based Reflex

Remembers recent info to fill in gaps it can't currently see.



03 Goal-Based

Makes decisions based on a clear goal it's trying to reach.



04 Utility-Based

Chooses the option with the highest expected value or usefulness.



05 Learning Agent

Gets better and smarter over time by learning from experience.



AI in Everyday Life

Recognizing rational agents you already use

IN PRACTICE



Voice Assistants

Siri and Alexa perceive speech (sensors) and respond via audio (actuators).



Recommendation Engines

Netflix models user preferences to improve relevance and engagement.



Self-Driving Systems

Operate in a partially observable, dynamic, multi-agent environment.



TRY IT YOURSELF

Describe an AI Using PEAS

- 1 In pairs, pick one AI: a shopping app, a warehouse robot, or a health-checkup app.
- 2 Write down its Performance, Environment, Actuators, and Sensors (PEAS).
- 3 Ask ChatGPT or Gemini to double-check your answer and suggest what you missed.
- 4 Be ready to explain your choice to the class.





■ TRY IT YOURSELF

What Type of Agent Is Your AI?

- 1 For the same AI from before, decide which intelligent-agent type fits it best.
- 2 Explain why: does it need memory? A goal? To learn over time?
- 3 Ask ChatGPT or Gemini if a smarter version would work better.
- 4 Compare your answer with another pair.

■ WRAPPING UP

Key Takeaways

- AI can be viewed in four ways — this course mainly uses the rational-agent view.
- AI has capability categories (ANI, AGI, ASI), broad branches, and two useful views: classical and modern AI.
- The Turing Test checks if you can tell a computer apart from a real human.
- PEAS is a simple 4-question checklist to describe any AI agent.
- AIMA describes five intelligent-agent types, from simple reflex agents to learning agents.
- AI has evolved from symbolic reasoning and expert systems to deep learning and today's generative AI.



Recommended Reading



Primary Text

Russell, S. & Norvig, P. (2010). Artificial Intelligence: A Modern Approach (3rd ed.). Chapters 1–2.



Reference Book

Norvig, P. (1992). Paradigms of Artificial Intelligence Programming: Case Studies in Common Lisp.



Further Practical Reading (optional)

Joshi, P. (2017). Artificial Intelligence with Python.

NEXT LECTURE

Knowledge Representation & Reasoning